

Econ 440: Motivation

Carlos Flores

Department of Economics
California Polytechnic State University

1 Motivation: Introduction

1.1 Two Main Uses of Econometrics

1. Prediction/Forecast
Ex: Forecast next year GDP, interest rates, sales, etc.
2. Get causal effects
Ex: Estimate the causal effect of attending college on earnings.

1.2 1 & 2 above are related.

- If X causes Y , X can be used to forecast Y .
- But, Z may be a better predictor of Y even if Z does not cause Y .
Ex: Using google trends to predict stock market, recessions, etc.
- Key question for 1: Do I predict (forecast) Y well?

$$Y = a + bZ + \epsilon$$

b : $\text{correlation}(Y, Z) = \text{corr}(Y, Z)$. Not a causal effect of Z on Y .

- Key question for 2: Are we really estimating a causal effect?

$$Y = a + bX + \epsilon$$

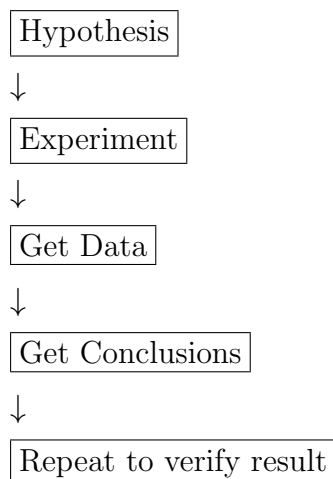
Is b the causal effect of x on y ? Say, does an increase in X by 1 unit causes an increase in Y by b ?

Our Focus in this Class: Estimation of Causal Effects.

2 Motivation: Causal Inference

- Economics is a social science.
- How does science advance? It uses the scientific method.

Scientific Method:



Simple example:

1. Hypothesis: Seeds need water to grow.
2. Experiment: Take 2 identical apple seeds, then you water one regularly, and the other do not. **Key:** Put both in otherwise exactly same conditions (e.g., same amount of soil, sunlight, wind, etc.), i.e., **ceteris paribus** $\implies$ Any difference would be due to watering, so we would get a causal effect of “watering” on “seed growing”.
3. Get data and conclusions.
4. Repeat for many more seeds (100’s), including different seeds (e.g., orange, grape, beans, etc.) *and* different conditions (e.g., windy, poor sunlight or soil conditions).

Economics Example:

1. Hypothesis: Attending college causes an increase in future earnings.

Super ideal experiment:

2. Take 2 clones (e.g. same race, gender, ability/IQ, family background, socioeconomic status, etc.) who have lived the exact same experiences up to age 18. Hence, a clone genetically, physically, emotionally, and psychologically $\implies$ ceteris paribus.

Then, put one of them in college and the other do not $\implies$ All constant except “college” (ceteris paribus) $\implies$ We would get the causal effect of “college” on “earnings”.

3. Get data and conclusions.
4. Repeat for many clones, including different races, genders, IQ’s, etc. and conditions (e.g., good/bad colleges, different majors, etc.).

Clearly: This experiment is not feasible.

Second best ideal experiment in such cases: **Randomized Experiments:**

Assign who attends college at random (say, by flipping a coin).

- General steps:

1. Hypothesis. Formulate our Hypothesis
2. Perform a Randomized Experiment
3. Get/Gather Data
4. Analyze Data
5. Estimation of Effects *and* Hypothesis Testing
6. Draw Conclusions

- Example:

Hypothesis: Attending college causes an increase in future earnings.

Randomized Experiment:

1. Take a random sample size n from the population (i.e. a sample for which all people in the population have the same probability of being in our sample) $\implies$ Our sample is representative of the population.
2. Randomly assign who attends college (e.g. by flipping a coin) and see their earnings after 10 years. If n_0 and n_1 are “large enough”, then the characteristics of the **treatment** and **control** groups will be equal on average (by the “law of large numbers”), e.g., same fraction of females/males, low/high IQ, rich/poor people, Whites, Hispanics, etc. $\implies$ All is “equal” or *ceteris paribus*, except for “college” (our *treatment*).

Get/Gather Data

Analyze Data

Estimation of Effect: Let α_0 be the average causal effect of attending college on earnings, and let y_i denote the earnings of individual i . Then, we estimate α_0 as:

$$\hat{\alpha} = \frac{\sum_{i=1}^{n_1} y_i}{n_1} - \frac{\sum_{i=1}^{n_0} y_i}{n_0} \quad (1)$$

In words, for the people in our sample, the average treatment effect equals the average earnings of those who attended college minus the earnings of those who did not.

Hypothesis Testing: Is α_0 statistically different from zero?

H_0 : $\alpha_0=0$

H_a : $\alpha_0 \neq 0$

Draw Conclusions

- In most economic applications, we may have a random sample (1 above) but not a randomized treatment assignment (2 above).
- Issue with using $\hat{\alpha}$ to estimate the average treatment effect α_0 of “college” (our *treatment*) on earnings (our *outcome*): “college” (our treatment) is not randomly assigned. Instead, those who attend college are usually, or on average:
 - Come from a higher socioeconomic status
 - Have higher ability/IQ
 - Are White, non-Hispanic
 - Are more motivated
 - Have no criminal history
 - Have no disabilities, etc.

$\implies$ Difference in average earnings between those with “college” and “no college” can be due to “college” OR, to any of those factors. This is called **confounding or selection problem**.

$\implies \hat{\alpha}$ no longer represents the causal effect of attending college on earnings.

Then, how can we draw causal conclusions (estimate causal effects) without an experiment?

- The “old” way (not so old; widely used in the 50’s, 60’s, 70’s, and even now in some fields):
 - OLS (Ordinary Least Squares): $Y_i = a + bT_i + cX_{1i} + dX_{2i} + eX_{3i} + \dots + \epsilon_i$
 - where T_i = “college or not” and the X ’s are observed characteristics such as gender, race, age, etc.
 - In general, is b the causal effect of T or Y? **No!**
- OLS gives association, NOT causation. Why? Two main issues:
 1. *Confounding or Selection Problem*: “ T ” (e.g., “college”) and “ Y ” (e.g., earnings) may still have common causes we are not controlling for.

$\implies$ b is likely not a causal effect. (In the context of Econ 339: there is “omitted variable bias”.)

2. *Reverse Causality Problem*: Is T causing Y , or is Y causing T ?

Example 1: Is “college” causing a person to have a higher socioeconomic status, or is having a high socioeconomic status causing an increase on the chances that a person attends college?

Example 2: What is the effect of the number of police officers on crime?

Hypothesis: Having more police officers will lead to lower crime rates.

One may estimate an OLS of the number of crimes on the number of police officers by, e.g., zip codes or cities adjusting for factors such as income, race, unemployment rate, etc. We usually find:

$$\text{corr}(\text{police}, \text{crime}) > 0$$

Some may conclude that police officers cause more crime, but this is just correlation! Why is this happening? It is actually high crime rates that cause cities to hire more police officers (Reverse Causality)!

Hence, we need to ask: Does T cause Y , or does Y cause T ?

$\implies$ In sum, we cannot rely on OLS to “magically” solve our causality problems. Then, what do we do?

In the last 20-30 years, there has been a “credibility revolution” in Empirical Economics.

- Some key names:
 - Don Rubin
 - Guido Imbens
 - James Heckman
 - David Card
 - Josh Angrist
 - Alan Krueger
 - Orley Ashenfelter

Key Questions:

1. What is a causal effect?
 - Use the “Potential Outcome Framework”, also called “Rubin Causal Model”
2. Under what conditions/assumptions can we estimate causal effects? Or, under what assumptions can we “jump” from association/correlation to causation?
 - We now use (in economics and other fields) different **Research Designs or Identification Strategies** (informally, tools/methods or approaches) to estimate causal effects.
 - Examples: Experimental methods, OLS (not too good), propensity score methods, instrumental variables (IVs), research discontinuity designs, difference-in-difference, bounds, etc.

Each method has different requirements, assumptions, advantages, and disadvantages.

Some Notes:

- Econometrics is not just “statistics”: Economic Theory (e.g., economic theory on education) and Institutional Knowledge (e.g., how the college/university system works in the US) play a key role in estimating causal effects in Economics:
 - To justify assumptions and enhance credibility of our estimates, i.e, to argue that we have estimated a causal effect and not just a correlation.
 - To come up with clever ways to estimate effects (e.g., see paper by Angrist and Pischke, 2010).

- The tools/methods that we will cover are also now widely used for “Program Evaluation”. Examples:
 - Do federally-funded job training programs help to increase earnings/employment?
 - What’s the effect of food stamps on health, employment?
 - What’s the effect of Medicaid on hospital use and health?
 - Do conditional cash programs (e.g. money given to poor families if kids attend school in poor countries) improve health and schooling? Does it help reduce poverty?
 - Do programs to help the poor in developing countries work?
 - Do Minimum Age Drinking Laws reduce alcohol consumption and car accidents?

Now, most governments and non-government organizations (e.g. World Bank) require evaluation of their programs for continuing funding. These methods are also used in the industry (e.g., in places like Amazon, Facebook, Google,) to estimate the effects of interventions (e.g., a new marketing campaign, product, or firm policy).

- For more, see the paper by Angrist and Pischke (2010).

2.1 Two General Lessons/Ideas to Take from This Class

1. *Correlation $\neq$ Causation*

- See above.
- $\text{Corr}(\text{education}, \text{earnings}) > 0$. It can be due to many factors: ability, high socioeconomic status (SES), etc.
- Suppose we want to get/estimate the causal effect of A on B, and we find that $\text{Corr}(A, B) > 0$ (or < 0). This could be due to:
 - (a) $A \rightarrow B$ (what we want)
 - (b) $B \rightarrow A$ (reverse causality)
 - (c) $C \rightarrow A \& B$ (confounding)

No way to tell without a clear research design!

We ask: under what assumptions can we jump from correlation to causation?

Avoid statements like those we typically hear on TV or radio: “A recent paper found that watching TV causes obesity”. Chances are, the paper found association, not causation! E.g., if that statement is true, then the implication is that we should ban TVs from all the gyms because they cause obesity (?). Of course not!

This is a trivial example, but unfortunately, that mistake is more common than you can imagine at all levels (industry, government, academia, etc.), and for very important topics!

2. No “Free Lunch” in Econometrics. Every research design/identification strategy has its pros and cons, even an experiment!

Experiment:

Pros:

- Get clear causal effect

Cons:

- Usually very costly
- Time consuming (e.g. for “college” example, we would need to wait many years to see outcomes; or the urgent need for a coronavirus vaccine).
- Ethical issues
- Individuals/“clones” may not do what we ask them to (e.g. some in control group may attend college anyways)

Avoid saying: “This approach is always perfect/adequate”.

Final Note:

- It is dangerous to jump from correlation to causation (see examples above, such as the police-crime or TV-obesity ones).

But: It may also be risky to ignore all associations/correlations: causality may be hidden in some, and it is not always easy to come up with good research designs or perform an experiment.

– Example: Effect of smoking on lung cancer:

→ In the past: All based on OLS-type analysis (1920’s, 30’s, 40’s)

→ Some people argued “correlation, not causation” e.g. Ronald Fisher, one of the fathers of randomized experiments

This is still an issue! Current examples: Effect of vaccines on autism? Effect of a possible vaccine for the coronavirus on the probability of “not getting” the disease, etc. It is hard getting causal effects!

Due to this, in general, sometimes causality is argued based on consistency of results (e.g., in the case of the effect of smoking on birth weight): find same/similar results across many studies using different approaches, data, populations, over time, control variables, etc. This may not be ideal, but sometimes it is the “only” option.

In summary: It is hard to do science!! In this class we will study how to estimate causal effects.